

Recursive Sequences: Problems, Solutions, and Answers

Problems 10, 14, 15, 16, 18, and 19

Problem 10: Limit of a nested-radical recurrence

Problem. Define $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \geq 1$. Find $\lim_{n \rightarrow \infty} a_n$.

Solution. We first show that the sequence is increasing and bounded above.

Since $a_1 = 1 < 2$, suppose $0 < a_n < 2$. Then

$$a_{n+1} = \sqrt{2 + a_n} < \sqrt{4} = 2.$$

Thus $a_n < 2$ for every n . Also, for $0 < a_n < 2$,

$$\begin{aligned} a_{n+1} > a_n &\Leftrightarrow \sqrt{2 + a_n} > a_n \Leftrightarrow 2 + a_n > a_n^2 \\ &\Leftrightarrow (2 - a_n)(a_n + 1) > 0, \end{aligned}$$

which is true. Therefore, (a_n) is increasing and bounded above, so it converges.

Let its limit be L . Passing to the limit in the recurrence gives

$$L = \sqrt{2 + L}.$$

Squaring,

$$L^2 - L - 2 = 0 \Rightarrow (L - 2)(L + 1) = 0.$$

Because every term is positive, L cannot be -1 . Hence $L = 2$.

Answer: $\lim_{n \rightarrow \infty} a_n = 2$

Problem 15: A reciprocal substitution

Problem. A sequence satisfies $a_1 = 1$ and $a_{n+1} = a_n/(1 + n a_n)$ for $n \geq 1$. Find a_{100} .

Solution. Take reciprocals. From the recurrence,

$$1/a_{n+1} = (1 + n a_n)/a_n = 1/a_n + n.$$

Define $b_n = 1/a_n$. Then

$$b_{n+1} = b_n + n, \quad b_1 = 1.$$

Therefore,

$$\begin{aligned} b_n &= 1 + (1 + 2 + \cdots + (n - 1)) \\ &= 1 + n(n - 1)/2. \end{aligned}$$

For $n = 100$,

$$b_{100} = 1 + (100 \cdot 99)/2 = 1 + 4950 = 4951.$$

Since $a_{100} = 1/b_{100}$,

Answer: $a_{100} = 1/4951$

Problem 14: Iteration modulo 9

Problem. Let $a_0 = 3$, $a_1 = 3$, and $a_n = a_{n-1}^2 - 2$ for $n \geq 2$. Find $a_{100} \pmod{9}$.

Solution. Compute the recurrence modulo 9:

$$a_2 \equiv 3^2 - 2 \equiv 7 \pmod{9},$$

$$a_3 \equiv 7^2 - 2 \equiv 49 - 2 \equiv 47 \equiv 2 \pmod{9}.$$

Now 2 is a fixed point of the recurrence modulo 9, because

$$2^2 - 2 = 2.$$

Thus $a_n \equiv 2 \pmod{9}$ for every $n \geq 3$. In particular,

Answer: $a_{100} \equiv 2 \pmod{9}$

Problem 16: Consistency check of the printed recurrence

Problem (as printed). Define $a_1 = 19$, $a_2 = 98$, and $a_{n+2} = a_n - 1/a_{n+1}$ for $n \geq 1$. Show that $a_n > 0$ for all n , and find $\lfloor a_{1999} \rfloor$.

Solution. As printed, the problem is internally inconsistent. Multiply the recurrence by a_{n+1} :

$$a_{n+1}a_{n+2} = a_n a_{n+1} - 1.$$

Let $b_n = a_n a_{n+1}$. Then

$$b_{n+1} = b_n - 1.$$

Since $b_1 = a_1 a_2 = 19 \cdot 98 = 1862$, we obtain

$$b_n = 1862 - (n - 1) = 1863 - n.$$

At $n = 1863$, this gives

$$a_{1863}a_{1864} = b_{1863} = 0.$$

Therefore at least one of a_{1863} and a_{1864} equals 0, contradicting the instruction to prove that $a_n > 0$ for all n .

Moreover, the recurrence then involves division by zero and cannot be continued to a_{1999} .

Conclusion. There is almost certainly a typographical error in either the initial value, the sign in the recurrence, or the requested index. The printed data do not determine a valid value of $\lfloor a_{1999} \rfloor$.

Answer as printed: no valid value; the problem contains a contradiction

Problem 18: Double-angle recurrence and exact period 3

Problem. A sequence is defined by $a_1 = \cos \theta$ and $a_{n+1} = 2a_n^2 - 1$ for a fixed $\theta \in (0, \pi)$. Find a_n in terms of θ , and determine all θ for which the sequence has period exactly 3.

Solution. Use the double-angle identity

$$\cos(2x) = 2\cos^2 x - 1.$$

Since $a_1 = \cos \theta$, induction gives

$$a_n = \cos(2^{n-1}\theta).$$

For the sequence to have period dividing 3, we need $a_4 = a_1$. Because the recurrence is deterministic, this equality makes all later terms repeat every three steps. Thus

$$\cos(8\theta) = \cos \theta.$$

The equation $\cos A = \cos B$ is equivalent to $A = 2\pi k \pm B$. Hence either

$$8\theta = 2\pi k + \theta \Rightarrow \theta = 2\pi k/7,$$

or

$$8\theta = 2\pi k - \theta \Rightarrow \theta = 2\pi k/9.$$

Restricting to $0 < \theta < \pi$ gives

$$\theta \in \{2\pi/7, 4\pi/7, 6\pi/7, 2\pi/9, 4\pi/9, 2\pi/3, 8\pi/9\}.$$

Because 3 is prime, a period dividing 3 is either 1 or 3. Period 1 occurs when $a_2 = a_1$:

$$\cos(2\theta) = \cos \theta.$$

In $(0, \pi)$, this gives $\theta = 2\pi/3$. We must exclude this value. Therefore the exact-period-3 values are

$$\text{Answer: } a_n = \cos(2^{n-1}\theta), \text{ and } \theta \in \{2\pi/7, 4\pi/7, 6\pi/7, 2\pi/9, 4\pi/9, 8\pi/9\}$$

Problem 19: A Fibonacci sequence hidden in a nonlinear recurrence

Problem. Let $a_1, a_2, \dots$ be positive real numbers with $a_1 = 1$, $a_2 = 2$, and $a_{n+2} = (a_{n+1}^2 + 1)/a_n$ for $n \geq 1$. Prove that every term is a positive integer, and find the largest $n \leq 100$ for which $a_n < 1000$.

Solution. The first few terms are

$$a_1 = 1, a_2 = 2, a_3 = 5, a_4 = 13, a_5 = 34, \dots$$

These are the odd-indexed Fibonacci numbers. Let $F_1 = F_2 = 1$ and $F_{m+2} = F_{m+1} + F_m$. We claim

$$a_n = F_{2n-1}.$$

We use the Fibonacci identity

$$F_{m+2}F_{m-2} = F_m^2 + (-1)^{m-1}.$$

When m is odd, this becomes

$$F_{m+2}F_{m-2} = F_m^2 + 1.$$

The formula is true for $n = 1$ and $n = 2$: $a_1 = F_1 = 1$ and $a_2 = F_3 = 2$. Suppose $a_n = F_{2n-1}$ and $a_{n+1} = F_{2n+1}$. Taking $m = 2n + 1$, which is odd, gives

$$F_{2n+3}F_{2n-1} = F_{2n+1}^2 + 1.$$

Therefore,

$$\begin{aligned} a_{n+2} &= (a_{n+1}^2 + 1)/a_n \\ &= (F_{2n+1}^2 + 1)/F_{2n-1} = F_{2n+3}. \end{aligned}$$

This proves by induction that $a_n = F_{2n-1}$ for every n . Hence every term is a positive integer.

Now compare consecutive relevant Fibonacci numbers:

$$a_8 = F_{15} = 610 < 1000,$$

$$a_9 = F_{17} = 1597 > 1000.$$

Since the Fibonacci numbers are increasing, the largest $n \leq 100$ satisfying $a_n < 1000$ is $n = 8$.

Answer: every a_n is a positive integer, $a_n = F_{2n-1}$, and the largest n is 8